AMLD+0 Settlement of the Geodesic Curvature-Profile Conjecture

Counterexample, Infinite Failure Family, and Curvature–Potential Criterion

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Abstract

We evaluate whether a minimum-edge verifier-legal directed proof path must minimize cumulative negative Ollivier–Ricci curvature burden. A six-vertex example refutes the universal claim: the unique two-edge geodesic incurs burden 1/3, while a three-edge detour incurs zero. We reproduce exact primal and dual optimal-transport certificates, extend the failure to a leaf-amplified infinite family, and retain the bound $0 \le B_-(P) \le 2|P|$. A calibrated potential condition is sufficient for a geodesic to minimize curvature burden. We also provide an explicit proof-chain protocol for Horn realization, distinguish edge length from curvature burden, and state the limits of the graph-theoretic results. No proof-search performance gain follows from the geometric comparison alone.

1 Frozen definitions and scope

Fix a connected finite simple undirected graph $G_u = (V, E_u)$ and an orientation selecting permitted directed moves $E_\to \subseteq \{(x, y) : \{x, y\} \in E_u\}$. Legal paths use only $E_\to$; Ollivier–Ricci curvature uses the underlying graph G_u. Let d_u be its unweighted shortest-path metric and m_x the uniform probability measure on the *distinct neighbors* $N_u(x)$, assuming no isolated endpoint of a curvature-tested edge. For adjacent x, y put

$$\kappa(x, y) = 1 - W_1^{d_u}(m_x, m_y), \qquad b(x, y) = \max\{0, -\kappa(x, y)\}.$$

For a finite legal directed path $P = (x_0, \dots, x_n)$ define

$$B_-(P) = \sum_{i=0}^{n-1} b(x_i, x_{i+1}), \qquad |P| = n.$$

Here a geodesic minimizes $|P|$ among legal paths with the given endpoints, *not* $B_-(P)$. The conjecture under assessment asserted, without supplementary hypotheses,

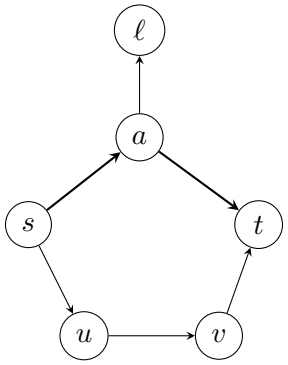
$$P^* \text{ is a directed } s\text{–}t \text{ geodesic} \quad \Longrightarrow \quad B_-(P^*) \le B_-(P) \text{ for every legal } P : s \to t.$$

2 Six-vertex counterexample and its exact certificates

Let $V = \{s, a, t, u, v, \ell\}$. The undirected edges are

$$\{s, a\}, \{a, t\}, \{t, v\}, \{v, u\}, \{u, s\}, \{a, \ell\},$$

and the legal directed edges are $s \to a \to t$, $s \to u \to v \to t$, and $a \to \ell$.



The unique two-edge legal s–t path is $P_g = (s, a, t)$; the detour is $P_d = (s, u, v, t)$ of length three. Thus P_g is the unique geodesic.

For edge $s - a$, the neighborhood measures are

$$m_s = \tfrac{1}{2}(\delta_a + \delta_u), \qquad m_a = \tfrac{1}{3}(\delta_s + \delta_t + \delta_\ell).$$

A primal feasible coupling sends $1/3$ of mass $a \to \ell$, $1/6$ of $a \to t$, $1/3$ of $u \to s$, and $1/6$ of $u \to t$. The total cost is

$$\tfrac{1}{3} \cdot 1 + \tfrac{1}{6} \cdot 1 + \tfrac{1}{3} \cdot 1 + \tfrac{1}{6} \cdot 2 = \tfrac{7}{6}.$$

For the dual, set $f(u) = 1$, $f(s) = f(a) = f(v) = 0$, and $f(t) = f(\ell) = -1$. Every graph edge changes f by at most one, so f is 1-Lipschitz. Its Kantorovich–Rubinstein value is

$$\int f \, dm_s - \int f \, dm_a = \tfrac{1}{2} - (-\tfrac{2}{3}) = \tfrac{7}{6}.$$

Primal and dual equality certify $W_1(m_s, m_a) = 7/6$, hence $\kappa(s, a) = -1/6$. Reflection of the five-cycle about the leaf-bearing vertex a exchanges s and t, giving $\kappa(a, t) = -1/6$. For detour edges $s{-}u, u{-}v, v{-}t$, supports of the two neighbor measures have no common vertex, so all transport must cost at least one per mass unit. For $s - u$, match $a \mapsto s$ and $u \mapsto v$; for $u - v$, match $s \mapsto u$ and $v \mapsto t$; for $v - t$, match $u \mapsto v$ and $t \mapsto a$. All six matched pairs are adjacent, and every involved neighbor distribution is uniform on two vertices. These explicit couplings cost exactly one, giving $W_1 = 1$ and $\kappa = 0$ for each detour edge. Consequently

$$\boxed{|P_g| = 2 < 3 = |P_d|, \qquad B_-(P_g) = \tfrac{1}{3} > 0 = B_-(P_d).}$$

This refutes the stated universal graph-theoretic conjecture.

Remark 2.1 (Explicit verifier-legal realization)**.** To avoid assuming that every abstract directed graph automatically arises as a genuine theorem graph, choose propositional atoms p_x for each $x \in V$, frozen assumptions $\Gamma = \{p_s\} \cup \{p_x \to p_y : (x, y) \in E_{\to}\}$, and a fixed protocol allowing one edge transition precisely by applying the indicated Horn implication to its antecedent. Every directed path corresponds to a legal sequence of independently checkable modus-ponens steps from p_s toward p_t. Here a protocol state tracks one current active atom p_x and the unchanged finite axiom table, rather than the full monotonically accumulated consequence BANK. A permitted step checks one recorded implication $p_x \to p_y$ and its single modus-ponens application, updates the active atom to p_y, and charges one edge; all checked implications remain frozen. This implements the directed *path protocol* used in the theorem. A general proof calculus might combine previous discoveries or use macros to add new transitions, so the conclusion is not asserted for its unrestricted proof-state graph.

3 Infinite leaf-amplified failure family

For each integer $r \geq 2$, replace ℓ with leaves $\ell_1, \ldots, \ell_r$ adjacent only to a, and direct the new leaf edges outward from a. Set $q = r + 2$. At $s - a$, source masses remain $m_s = \frac{1}{2}(\delta_a + \delta_u)$ and target demands are $1/q$ at each of $s, t, \ell_1, \ldots, \ell_r$. For $r \geq 2$, send $1/2$ mass from a to the leaves (distance one); send $1/q$ from u to s (distance one), $1/q$ to t (distance two), and the remainder $1/2 - 2/q$ to leaves (distance three). Its cost is

$$\tfrac{1}{2} + \tfrac{1}{q} + \tfrac{2}{q} + 3(\tfrac{1}{2} - \tfrac{2}{q}) = 2 - \tfrac{3}{q}.$$

A matching dual potential is

$$f(u) = 1, \quad f(s) = f(v) = 0, \quad f(a) = f(t) = -1, \quad f(\ell_j) = -2 \quad (1 \leq j \leq r).$$

It is 1-Lipschitz on the full undirected graph and its dual objective is $2 - 3/q$. Thus

$$\kappa(s, a) = \kappa(a, t) = -\frac{r-1}{r+2}, \quad B_-(P_g) = \frac{2(r-1)}{r+2}, \quad B_-(P_d) = 0, \quad \lim_{r\to\infty} B_-(P_g) = 2.$$

The $r = 1$ six-vertex example was computed separately above; the displayed $r \geq 2$ coupling must not be extrapolated to $r = 1$. Adding the leaves changes local neighborhood geometry while leaving both directed route lengths fixed, establishing the off-path neighborhood obstruction.

4 Universal curvature-burden bound

Theorem 4.1. *For every adjacent pair x, y in an unweighted undirected graph with neighbor measures as above, $-2 \leq \kappa(x, y) \leq 1$. Every n-edge verifier-legal path therefore satisfies $0 \leq B_-(P) \leq 2n$, and every directed geodesic P_g from s to t satisfies $B_-(P_g) \leq 2d_\to(s, t)$.*

Proof. For $z \in N_u(x)$ and $w \in N_u(y)$, the walk $z - x - y - w$ has length at most three. Hence all couplings between m_x and m_y cost at most three and $0 \leq W_1 \leq 3$. Rearrangement gives the curvature bound; summing nonnegative edge burdens gives $B_- \leq 2n$. For a directed geodesic $n = d_\to(s, t)$. □

5 A curvature–potential bridge, with its exact scope

Theorem 5.1 (Curvature–Potential Geodesic Bridge)**.** *Fix reachable vertices s, t. Suppose there exist $\alpha \geq 0$ and a function $\phi : V \to \mathbb{R}$ such that every legal directed edge satisfies*

$$b(x, y) \geq \alpha + \phi(x) - \phi(y).$$

If a directed length geodesic *P_g from s to t attains equality on* each edge*, then P_g minimizes total negative-curvature burden among all legal s–t paths.*

Proof. For any legal $P : s \to t$ telescope the edge inequalities to get $B_-(P) \geq \alpha|P| + \phi(s) - \phi(t)$. Since $|P| \geq |P_g|$ and $\alpha \geq 0$, this is at least $\alpha|P_g| + \phi(s) - \phi(t)$. Equality on all edges of P_g makes the latter expression $B_-(P_g)$. □

Remark 5.2. The condition permits $\alpha = 0$. The existence of a calibrated potential saturating a chosen geodesic is a *substantive additional hypothesis*, not inferred from geodesicity or curvature. The bridge is an optimality certificate of a familiar potential/telescoping kind, not an algorithm for constructing ϕ, and does not imply the proposed AMLD search policy will discover the route.

6 Interpretation, reproducible checks, and boundaries

Local curvature is not determined by vertex degree alone: an edge of a triangle K_3 has $\kappa = 1/2$, while a cycle edge of C_5 has $\kappa = 0$, although both examples have degree-two endpoints and unit edge length. Graph curvature describes local transportation geometry; legal proof distance is global and directed.

The displayed exact transport couplings and dual functions constitute mathematical optimality certificates. Additional exact rational min-cost-flow computations agreed with the formulas for $r = 1, \ldots, 12$, but finite checks do not replace the symbolic proof. No claim is made that six vertices is the minimum possible number for a counterexample; that would require a separate exhaustive argument. A topological graph may serve as a proof-step protocol only after its allowed edges and verification receipts have been fixed. The construction does not apply automatically to unrestricted proof calculi.

Settlement classification. The universal strong conjecture is REFUTED in the graph and the specified Horn step protocol. Theorems 4.1 and 5.1 remain conditional positive results. Whether curvature profiles statistically predict short or useful proofs in a real prover is untested here. The stated mathematical counterexample does not imply that six independent AMLD or FORGE-6 agents have executed the corresponding search.

Scope. The infinite-family coupling is stated for $r \geq 2$, the potential bridge requires edgewise saturation along the chosen geodesic, and the Horn realization concerns the declared path protocol, not every possible proof-search policy.